AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

General Mathematics Paper I

Time allowed: 40 minutes Marks 30

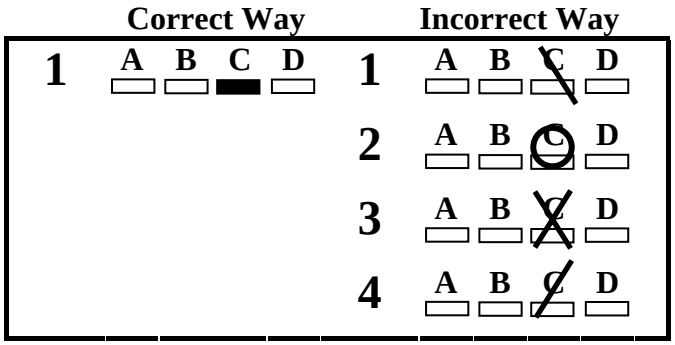
INSTRUCTIONS

1. Read each question carefully.

2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.

3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.

4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the box for your choice with a pencil as shown below.



5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new box.

6. DO NOT write anything in the answer grid. The computer only records what is in the boxes.

7. You may use a simple calculator if you wish.

1. The price of a toy car is Rs. 100 and reduced to 50% during Eid season. After Eid, the price of the car is increased by 50%. The final price of the car is

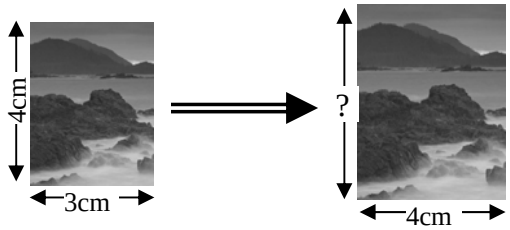
   A. Rs. 25
   B. Rs. 50
   C. Rs. 75
   D. Rs. 100

2. In a two hours workshop, 20 minutes were given for tea break. The ratio of the break time to the total time of the workshop is

   A. 1 : 10
   B. 10 : 1
   C. 6 : 1
   D. 1 : 6

3. A photographer has proportionally enlarged a photograph of dimensions $3\text{cm} \times 4\text{cm}$. After enlargement if the width is 4cm, then the length is

   A. 3cm
   B. 3.33cm
   C. 5cm
   D. 5.33cm



4. After the death of Mr. Saleem, his savings were divided between his daughter and son according to Islamic laws. The son gets twice as that of the daughter. If the daughter gets Rs. 2,00,000, then the total saving of Mr. Saleem is

   A. Rs. 3,00,000
   B. Rs. 4,00,000
   C. Rs. 6,00,000
   D. Rs. 9,00,000

5. Mr. Amjad earned Rs. 5,00,000 from a mango farm and Rs. 3,00,000 from a banana farm. The amount of *Ushr* he has to pay is [Note: Rate of *Ushr* is 10% of earning]

   A. Rs. 20,000
   B. Rs. 30,000
   C. Rs. 50,000
   D. Rs. 80,000

6. If the cost price of a handicraft is Rs. 135 and it is sold at a loss of Rs. 25, then its selling price is

   A. Rs. 160
   B. Rs. 150
   C. Rs. 110
   D. Rs. 90

7. A customer paid Rs.100 for a book after getting 20% discount. What was the tag price of book?

   A. Rs. 125
   B. Rs. 120
   C. Rs. 115
   D. Rs. 80

8. A merchant gained 25% profit by selling a vase for Rs.175. What is the cost of the vase?

   A. Rs. 135
   B. Rs. 140
   C. Rs. 145
   D. Rs. 150

9. Sadia spent 1500 Saudi Riyals during her visit to Saudi Arabia. If one dollar is equal to 3.1 Saudi Riyals, then the amount she spent in dollars was

   A. $483.8
   B. $497.8
   C. $513.8
   D. $526.8

10. If the principal amount is Rs. 2,000 and the rate of interest is 2% per annum, then after 1 year the markup will be

   A. Rs. 20
   B. Rs. 40
   C. Rs. 200
   D. Rs. 400

11. Through a debit card, a customer can

   A. lend cash from bank.
   B. make payments to the bank.
   C. deposit money in his bank account.
   D. make payments from his bank account.

12. A company sells motorcycles on leasing. The cost of a motorcycle is Rs. 45,000 and according to the leasing plan the motorcycle is delivered on 30% down payment and 10 equal monthly instalments. Each instalment is of

   A. Rs. 4,500
   B. Rs. 3,500
   C. Rs. 3,150
   D. Rs. 3,000

13. At an electronics appliance shop, the marked price of a television is Rs. 17,500. If General Sales Tax (GST) is levied at a rate of 16%, then the total cost of the television for a buyer is

   A. Rs. 2,800
   B. Rs. 14,700
   C. Rs. 17,516
   D. Rs. 20,300

14. Mr. Wajid paid an income tax of Rs. 1400 in June. If his gross income for the month is Rs. 40,000, then the rate of income tax is

   A. 0.035%
   B. 0.35%
   C. 3.5%
   D. 35%

15. In the given table, cost of 1 unit of electricity is given for different ranges of units consumed.

Units consumed	1 – 100	101 – 200	201 – 400	400 or more
Cost of 1 unit (in Rs.)	7	11	20	25

In the month of January, Mr. Anwar's family consumed 305 units of electricity. The cost of electricity for the month of January, excluding all the taxes is

A. Rs. 3,355
B. Rs. 3,900
C. Rs. 6,100
D. Rs. 7,625

16. A cellular company pays Rs. 200 per hour plus commission of Rs. 30 on sale of each SIM to its sales person. If a sales person sold 4 mobile SIMs in 3 hours, then the company will pay him

A. Rs. 320
B. Rs. 630
C. Rs. 720
D. Rs. 890

17. The expression $\sqrt[5]{x^5 y^5}$ is equal to

A. $x^{25} y^{25}$
B. $x^{10} y^{10}$
C. $x^5 y^5$
D. xy

18. If reciprocal of 2^{-3} is 2^n, then the value of n is

A. -3
B. -2
C. 2
D. 3

19. The radius of moon is 1738140m (approx). In scientific notation it is equal to

A. 1.73814×10^4 m.
B. 1.73814×10^5 m.
C. 1.73814×10^6 m.
D. 1.73814×10^7 m.

20. The arithmetic mean between $\sqrt{3}$ and $\sqrt{27}$ is

A. $2\sqrt{3}$
B. $4\sqrt{3}$
C. $\sqrt{15}$
D. $\frac{\sqrt{30}}{2}$

21. If a sequence is generalized as n^2+1, then the first term of the sequence is

A. 0
B. 1
C. 2
D. 3

22. The number of terms in the geometric progression 1, $3, 3^2$, 3^3,……., 3^6 is

A. 5
B. 6
C. 7
D. 8

23. If $A=\{0,-1,-2\}$ and $B=\{0,1,2\}$, then $A\cap B$ is

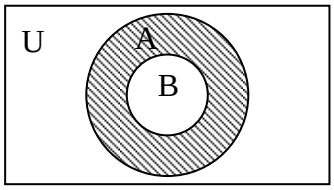
A. $\{\ \}$
B. $\{\phi\}$
C. $\{0\}$
D. $\{0,1,2\}$

24. For any three sets *A*, *B* and *C*, $A\cup(B\cup C)$ is equal to

A. $(A\cup B)\cap(A\cup C)$
B. $(A\cap B)\cup(A\cap C)$
C. $(A\cup B)\cup C$
D. $(A\cap B)\cap C$

25. The shaded area in the given Venn diagram represents

A. $A\cup B$
B. $A\cap B$
C. $B-A$
D. $A-B$

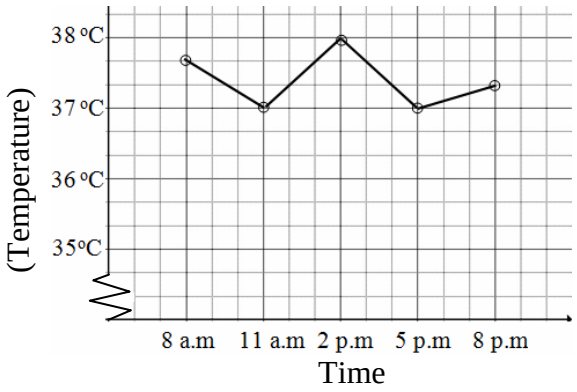


26. If $r=\{(1,a),(2,b)\}$ is a binary relation, then the domain of *r* is

A. $\{(1,2)\}$
B. $\{1,2\}$
C. $\{(a,b)\}$
D. $\{a,b\}$

27. The given graph was drawn by taking the temperatures of a patient on a certain day at different times. According to the graph, the temperature at 4pm is

A. 37 °C
B. 37.33 °C
C. 37.66 °C
D. 38 °C



28. A straight line $2x + y = 3$ passes through the point

A. $(0,1)$
B. $(1, 1)$
C. $(-1,0)$
D. $(1, -5)$

29. The ages, in years, of nine children in a group are 7, 7, 8, 9, 6, 8, 5, 8 and 10. The modal age is

A. 6
B. 7
C. 8
D. 9

30. Ahmed arranged a number of books owned by his friends in ascending order as 10, 12, 15, *x*, 17, 18. If the median of the data is 16, then the value of *x* is

A. 15
B. 15.5
C. 16
D. 17

END OF PAPER

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